
Ultrasonic Tissue Characterization of Uveal Melanoma and Prediction of Patient Survival After Enucleation and Brachytherapy

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We performed survival studies on 136 patients with uveal malignant melanoma who were examined with ultrasonic tissue characterization before treatment with cobalt-60 brachytherapy (74 patients) or enucleation (62 patients). Mean follow-up time was 58.7 months for the brachytherapy group and 59.0 months for the enucleated group. The maximal follow-up time was nearly ten years. Univariate survival analysis showed that patients with small tumors ($< 250 \text{ mm}^3$, 49 patients) had a higher five-year survival when treated with brachytherapy than when treated with enucleation. No patients in this study with tumor volumes greater than $1,500 \text{ mm}^3$ (13 patients) were treated with brachytherapy. For tumors of intermediate size (74 patients), survival analysis did not indicate appreciable differences between the treatment options. However, multivariate models including two ultrasonic tissue characterization variables, scatterer size and concentration, showed appreciable intergroup differences in the dependence of survival on these factors. Results

suggest that tissue properties detectable with ultrasonic techniques are related to differences in patient survival and may be used for treatment planning for tumors of intermediate size.

THE ISSUE OF RATIONAL TREATMENT for uveal melanoma (whether to treat with radiotherapy and attempt to preserve vision or to enucleate) is analytically difficult. The difficulties include the rarity of the tumor (approximately 1,500 new cases per year in the United States) and the long duration of follow-up needed to draw statistical inferences about treatment options (mean patient survival duration is over five years). Clinical studies must also consider the influence of known covariate factors such as tumor size and location, patient characteristics, and cell morphometry, as well as the possibility of unknown forms of bias.

Retrospective clinical studies comparing survival rates of patients whose eyes were enucleated and patients treated with radiotherapy are equivocal or indicate no marked difference in long-term survival for patient cohorts as a whole.¹⁻⁶

Histopathologic and micromorphometric studies of tissue from enucleated eyes have indicated that the underlying morphologic characteristics of tissue and mitotic state correlate with patient survival and may be used *ex post facto* to provide an estimate of survival.⁷⁻¹² These studies show that survival is influenced, in quantifiable terms, by the underlying morphologic characteristics of the tumor.

Previously, we demonstrated that uveal malignant melanoma can be differentiated into acoustic subgroups using ultrasonic tissue characterization techniques and that tissue characteristics correlate with morphometric

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and micromorphometric features in enucleated cases.^{13,14} We also showed that a statistical model comprising tumor volume, location, and acoustic scatterer concentration can be used to predict survival after enucleation.¹⁵ In this study we extend this model to compare survival of patients treated with radiotherapy (cobalt-60 brachytherapy) or enucleation and develop a model to determine survival estimates for individual classes of patients.

Patients and Methods

Ultrasonic tissue characterization became available for the evaluation of ocular tumors seen at New York Hospital-Cornell Medical Center as early as 1979. Since then, 1,375 scan sessions have been performed on 570 patients with tumors of all kinds. Follow-up data were obtainable on 62 patients with uveal melanoma whose eyes were enucleated. One hundred forty patients were treated with cobalt-60 brachytherapy. Pretreatment scans and posttreatment follow-up data were obtained on 74 of these patients (of the remaining patients, ultrasonic tissue characterization data were not obtained on 55 patients before treatment, four patients did not have adequate follow-up, and seven were excluded because cobalt-60 plaque was used in combination with other treatment modalities). Patient survival subsequent to treatment was determined from medical records, contact with patients or their families, and death certificates. Extensive skip tracing was used to obtain well-documented follow-up.

Before treatment with cobalt-60 plaque or enucleation, patients were scanned by diagnostic ultrasonography using the immersion technique. Tumor height and base dimensions were determined during the ultrasonic examination. The ultrasonic system operated at a center frequency of 10 MHz with a 15-dB band width of 10 MHz. Data were digitized at a sample rate of either 50 or 100 MHz.

Tissue characterization was performed by the normalized power spectrum method. Mean acoustic scatterer diameter and CQ^2 (the product of scatterer concentration, C ; and the square of relative impedance, Q) were then computed from the slope (dB/MHz) and 0-MHz intercept (dB) of the linear best fit of the power spectrum.¹⁴ On the basis of the simplifying assumption that relative impedance is close to unity, CQ^2 will generally be referred to as acoustic

concentration. The value of CQ^2 is presented in decibel form ($10 \log[CQ^2]$) in units of scatterers per cubic millimeter. Estimated tumor volumes were computed from tumor base and height dimensions using an ellipsoidal solid model.¹⁶

The relationship between clinical factors, ultrasonic tissue characteristics, and patient survival were examined using univariate and multivariate statistical techniques. Kaplan-Meier survival analysis, a univariate technique, was used for comparison of study subpopulations. The Cox multiple risk model,¹⁷ a multivariate technique for discrete and continuous variables, was used for assessment of risk factors and their interdependencies. A generalization of the Cox model, the Andersen-Gill model,¹⁸ was used along with graphic analysis of residuals to examine model validity. Multiple regression analysis, where the logarithm of survival time is a linear function of the independent variables, was used for the predictive model.¹⁹ In these analyses, patient deaths caused by metastatic disease were considered terminal. All others, including patient deaths of other or indeterminate causes, were considered censored.

The models assume radiation doses applied to individual patient's tumors are therapeutically equivalent. Target apex dose from the cobalt-60 applicator was 8,000 to 10,000 mGy. However, total tumor dose varied according to different tumor sizes, changing plaque activity, and application time.

Results

Survival statistics for the patient population stratified by treatment, tumor volume, location, and backscatter characteristics were analyzed (Table 1). The results indicated that survival decreased with tumor size and with ciliary body involvement. The results also suggested poorer survival for patients treated with enucleation than for patients treated with brachytherapy, but this will later be shown to have been caused by the covariate factor of tumor size.

The study population was stratified by volume for each treatment. Kaplan-Meier survival statistics (Table 2) suggested that patients with small tumors ($<250 \text{ mm}^3$) who were treated with brachytherapy had a higher survival rate than those whose eyes were enucleated (92% vs 74%, five-year survival), although statistically

TABLE 1
AGGREGATE SURVIVAL STATISTICS

FACTOR	NO. OF PATIENTS	DIED	FIVE-YEAR SURVIVAL (%)	SCORE	D	P VALUE
Volume (mm ³)						
< 250	49	8	85.8	11.5	—	—
250-750	50	14	75.9	-1.3	—	—
750-1,500	24	8	77.3	2.8	—	—
> 1,500	13	7	34.1	-43.6	12.44	.006
Location						
Posterior	111	27	81.4	5.4	—	—
Ciliary body	25	10	50.3	-23.8	6.83	.009
Treatment						
Enucleation	62	22	69.8	-8.7	—	—
Brachytherapy	74	15	81.5	7.3	3.41	.065
10Log scatterer concentration						
< -14	69	19	76.1	-1.25	—	—
> -14	67	18	75.9	1.28	0.09	.769
Scatterer diameter (μm)						
< 70	67	22	78.7	-1.24	—	—
> 70	69	15	73.0	1.20	0.08	.777

this was only moderately significant ($P = .113$). For patients with intermediate-sized lesions (250 to 750 mm³ and 750 to 1,500 mm³), no significant differences in survival were found between the two treatments. Interestingly, the five-year survival for patients with enucleated eyes was nearly constant for tumors having volumes of less than 1,500 mm³, whereas the percentiles for patients treated with radiotherapy decreased from 92% to 75% to 60% for tumor volumes less than 250 mm³, 250 to 750 mm³, and 750 to 1,500 mm³, respectively. Brachytherapy was not used in patients with tumors greater than 1,500 mm³ in size (the largest group). The five-year survival percentile for the patients with enucleated eyes in this group was 34%.

Because radiotherapy appeared to be effective in terms of survival for tumors less than 250 mm³ in size and had the added benefit of preserving visual function, we restricted the remainder of our analysis to the group of patients with intermediate-sized tumors (250 to 1,500 mm³).

A Cox model was developed for intermediate-sized tumors for each treatment group on the basis of tissue scatterer characteristics and tumor location. This multivariate technique assessed the risk of each possible factor, while taking into account the effects of other cofactors (Table 3). The results of this analysis indicate that risk is related primarily to the diameter of scatterer and tumor location for the subgroup of patients with enucleated eyes. For the brachy-

TABLE 2
COMPARISON OF SURVIVAL RATES FOR ENUCLEATION AND BRACHYTHERAPY WITH
STRATIFICATION BY TUMOR VOLUME

GROUP (mm ³)	ENUCLEATION			BRACHYTHERAPY			P VALUE
	NO. OF PATIENTS	DIED	FIVE-YEAR SURVIVAL (%)	NO. OF PATIENTS	DIED	FIVE-YEAR SURVIVAL (%)	
< 250	17	5	74.0	32	3	92.2	.110
250-750	14	4	76.6	36	10	75.4	.635
750-1,500	18	6	82.2	6	2	60.0	.431
> 1,500	13	7	34.1	0	0	—	—

TABLE 3
COX MULTIVARIATE RISK MODEL

FACTOR	χ^2	P VALUE	R ²	β	STANDARD ERROR OF β
Enucleation					
Diameter	7.38	.0066	.243	-.3037	.1118
Concentration	0.07	.7880	.003	-.0238	.0088
Location	6.63	.0100	.224	1.978	.7661
Model	14.76	.0020	.391	—	—
Brachytherapy					
Diameter	3.07	.0799	.106	-.0878	.0502
Concentration	5.58	.0182	.177	-.2525	.1069
Location	5.32	.0211	.170	1.619	.7018
Model	10.91	.0122	.296	—	—

therapy-treated eyes, concentration of scatterer and tumor location were the most important risk factors. Both models were found to be statistically significant (enucleation, $P = .002$; brachytherapy, $P = .012$). Examination of residuals from the Cox and equivalent Andersen-Gill models showed no basic violations of model assumptions or strong leverage effects, thus confirming model adequacy and accuracy.

The data were further analyzed by use of a multiple regression model appropriate for censored data (Table 4). This is a predictive model in which the logarithm of survival time is a linear function of the risk factors of tumor location, scatterer diameter, and scatterer concentration. Survival rate was also calculated using this model (Figs. 1 through 4).

The multiple regression model permits computation of predicted patient survival for patients with tumors with calculated volumes of 250 to 1,500 mm³ for either treatment, given

tumor location and backscatter characteristics. The 95% certainty ranges for the survival rates were determined (Fig. 4). Uncertainty bounds for survival estimates for the alternate treatments show regions of overlap and nonoverlap for a specific value of acoustic backscatter properties. For patients with certain tumors, predicted survival would be statistically different for the alternate treatments; for others it would not. For example, a posteriorly located tumor with scatterer diameter of 69 μm and concentration of $10^{-0.96} \text{ mm}^{-3}$ would have a predicted survival of 147 months (range, 112 to 195 months) with enucleation and 132 months (range, 105 to 166 months) with brachytherapy, where the number in parentheses indicates the 95% confidence interval. For such a case, there does not appear to be any statistically meaningful difference in predicted survival for the alternative treatments. Cases in which no statistical difference in predicted survival exists between the alternative therapies would fall near the line of interception of the appropriate survival surfaces. In contrast, in a case with scatterer diameter of 59 μm and concentration of $10^{-0.80} \text{ mm}^{-3}$, the survival estimates are 32 months (range, 19 to 53 months), and 123 months (range, 91 to 166 months) for enucleation and brachytherapy, respectively, a statistically significant difference.

Discussion

We determined that differences in the survival rate of patients with intermediate-sized uveal melanomas who were treated with enu-

TABLE 4
MULTIVARIATE REGRESSION MODEL

FACTOR	β	STANDARD ERROR OF β	T-TEST	P VALUE	SEQUENTIAL R ²
Enucleation					
Y-intercept	-2.237	.5848	—	—	—
Diameter	.0682	.0088	7.27	<.0001	.317
Concentration	-.000154	.00958	-0.02	.9873	.326
Location	-.6414	.0801	-8.00	<.0001	.795
F = 36.17, P < .001					
Brachytherapy					
Y-intercept	1.847	.3262	—	—	—
Diameter	.0096	.0049	1.97	.0562	.007
Concentration	.0406	.0073	5.60	<.0001	.410
Location	-.1589	.0860	-1.85	.0723	.459
F = 10.75, P < .001					

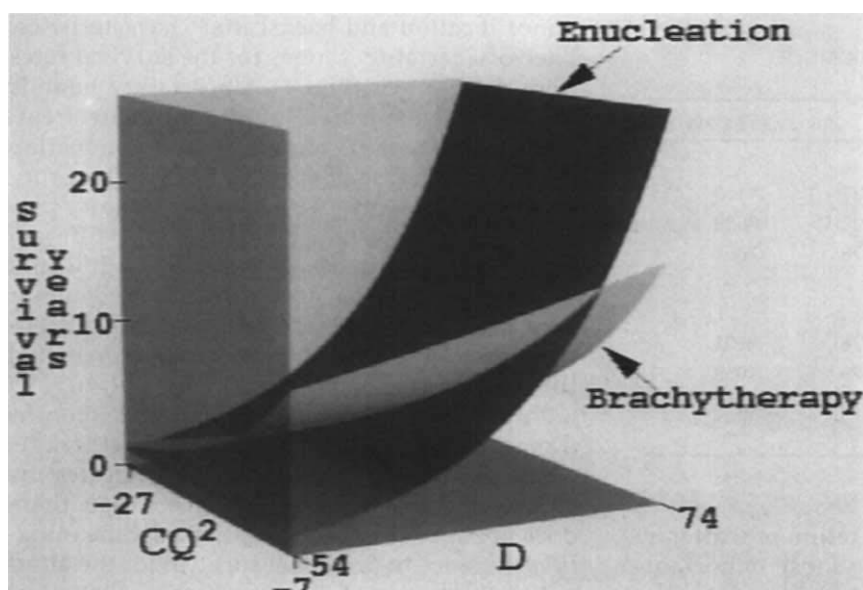


Fig. 1 (Coleman and associates). Contour plot representing predicted survival of enucleated vs brachytherapy-treated posterior melanomas. Survival (vertical axis) is a function of mean scatterer diameter and concentration, which together define the horizontal plane.

enucleation or Cobalt-60 brachytherapy can be modeled using a multivariate approach with the following quantifiable factors: (1) tumor acoustic scatterer characteristics, (2) tumor volume measurement, and (3) tumor location. The models demonstrate that differences exist in the response of uveal melanoma to alternate forms of treatment and that these differences can be identified by noninvasive means. The models were extended to obtain predictive equations with confidence intervals that can be used to compare estimated survival for the alternate treatments in individual cases.

The models' clinical factors, estimated tumor volume and location, have previously been shown to correlate with patient survival.⁶⁻⁸ The

acoustic scatterer factors have been shown to correlate to histopathologic and micromorphometric features of the tumor, as well as with patient survival in cases that required enucleation of eyes.^{13,15}

The acoustic backscatter factors used in the models are related to the spatial distribution, size, and density of tissue inhomogeneities, and are primarily sensitive to tissue structures ranging from about 25 to 75 μm in cross section. They may be considered analogous to statistical measures of cell distribution patterns as seen through an optical microscope, rather than assessments of the characteristics of individual cells.

Although it is possible that analysis of histo-

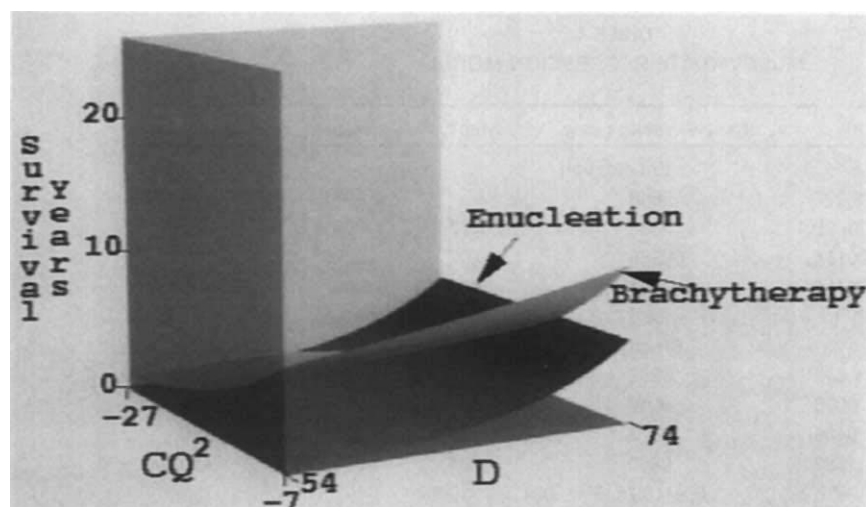


Fig. 2 (Coleman and associates). Contour plot representing predicted survival of enucleated vs brachytherapy-treated melanomas involving the ciliary body.

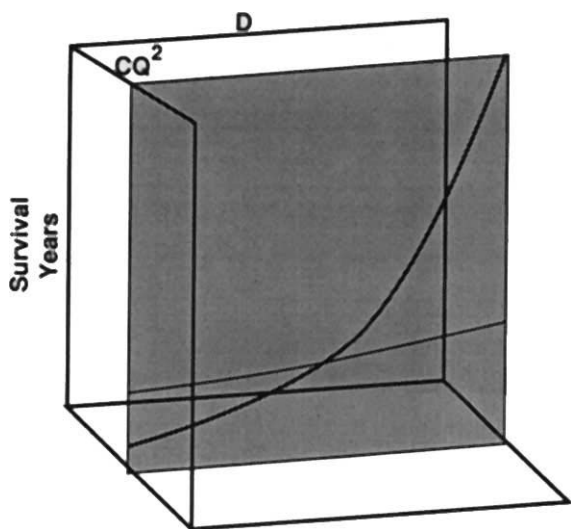


Fig. 3 (Coleman and associates). Orthogonal planes cutting across three-dimensional projections define survival curves as functions of either scatterer diameter or concentration.

pathologic characteristics could provide a statistically significant assessment of patient survival as is suggested by data from cases of enucleated eyes,¹⁵ it would require an invasive technique. Fine-needle biopsy procedures could be used if a sufficiently representative sample of the tumor would be obtained. However, at present, biopsy of intraocular tumors is generally regarded as unwarranted because of risks of tumor seeding or hemorrhage and is

reserved for situations in which the diagnosis is uncertain. The current model factors described in this study may be obtained noninvasively before treatment selection and can evaluate the entire tumor volume.

In this study, survival rates for patients with uveal melanomas of intermediate size were estimated. Statistically significant differences in survival were found between enucleated and brachytherapy-treated eyes for certain combinations of tumor acoustic characteristics, volume, and location. The usefulness and accuracy of the survival estimates were bound by the error associated with the model coefficients. Larger study populations would allow us to define model variance better and improve survival estimates. Also, any prospective use of the model for treatment planning would be predicated on correct initial diagnosis of uveal malignant melanoma. We currently use an ultrasonic tissue characterization diagnosis protocol to determine tumor type and subtype; however, our database does not include sufficient data to subclassify masquerading lesions such as leiomyoma (diktyoma) or melanocytoma.¹³ Additionally, the model cannot be used for previously treated tumors because treatment modifies tumor acoustic characteristics.

Our model using ultrasonic tissue characterization techniques provides a unique method for comparing the efficacies of different forms of melanoma treatment and may allow ocular melanoma treatment to be tailored to maximize individual patient survival.

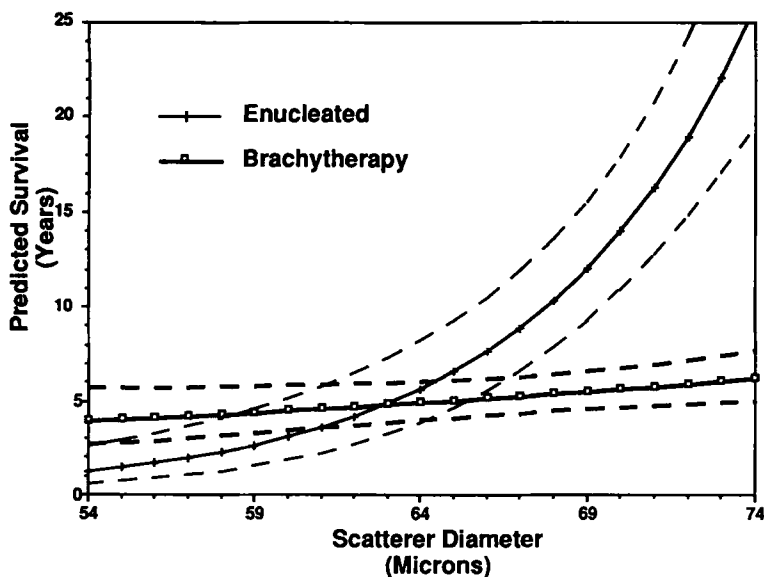


Fig. 4 (Coleman and associates). The plot demonstrates the relationship defined in Figure 1 for posteriorly located tumors having a scatterer concentration of -17 (the average for the study population) and includes the 95% confidence bounds. For each patient and each of the two possible treatment modalities, a measured scatterer concentration and diameter (combined with volume and location) will define a point (with known statistical uncertainty) within the three-dimensional projection.

References

1. Manschot, W. A., and van Strik, R.: Is irradiation a justifiable treatment of choroidal melanoma? Analysis of published results. *Br. J. Ophthalmol.* 71:348, 1987.
2. Gass, J. D. M.: Comparison of prognosis after enucleation vs cobalt-60 irradiation of melanomas. *Arch. Ophthalmol.* 103:916, 1985.
3. Markoe, A. M., Brady, L. W., Shields, J. A., Augsburger, J. J., Micaily, B., Damsker, J. J., Day, J. L., and Gamel, J. W.: Malignant melanoma of the eye. Treatment of posterior uveal lesions by Co-60 plaque radiotherapy versus enucleation. *Radiology* 156:801, 1985.
4. Augsburger, J. J., Gamel, J. W., Sardi, V. S., Greenberg, R. A., Shields, J. A., and Brady, L. W.: Enucleation versus cobalt plaque radiotherapy for malignant melanomas of the choroid and ciliary body. *Arch. Ophthalmol.* 104:655, 1986.
5. Augsburger, J. J., Gamel, J. W., Lauritzen, K., and Brady, L. W.: Cobalt-60 plaque radiotherapy vs enucleation for posterior uveal melanoma. *Am. J. Ophthalmol.* 109:585, 1990.
6. Seddon, J. M., Gragoudas, E. S., Egan, J. M., Glynn, R. J., Howard, S., Fante, R. G., and Albert, D. M.: Relative survival rates after alternative therapies for uveal melanoma. *Ophthalmology* 97:769, 1990.
7. Flocks, M., Gerende, J. H., and Zimmerman, L. E.: The size and shape of malignant melanomas of the ciliary body in relation to prognosis and histologic characteristics. A statistical study of 210 tumors. *Trans. Am. Acad. Ophthalmol. Otolaryngol.* 59:740, 1955.
8. McLean, I. W., Foster, W. D., and Zimmerman, L. E.: Prognostic factors in small malignant melanomas of the choroid and ciliary body. *Arch. Ophthalmol.* 95:48, 1977.
9. Weinhaus, R. S., Seddon, J. M., Albert, D. M., Gragoudas, E. S., and Robinson, N.: Prognostic factor study of survival after enucleation for juxtapapillary melanomas. *Arch. Ophthalmol.* 103:1673, 1985.
10. Gamel, J. W., McLean, I., Greenberg, R. A., Naidu, R. M., Folberg, R., Donoso, L. A., Seddon, J. M., and Albert, D. M.: Objective assessment of malignant potential of intraocular melanomas with standard microslides stained with hematoxylin-eosin. *Hum. Pathol.* 16:689, 1985.
11. Donoso, L. A., Augsburger, J. J., Shields, J. A., Greenberg, R. A., and Gamel, J.: Metastatic uveal melanoma. Correlation between survival time and cytomorphometry of primary tumors. *Arch. Ophthalmol.* 104:76, 1986.
12. Seddon, J. M., Polivogianis, L., Hsieh, C.-C., Albert, D. M., Gamel, J. W., and Gragoudas, E. S.: Death from uveal melanoma. Number of epithelioid cells and inverse SD of nucleolar area as prognostic factors. *Arch. Ophthalmol.* 105:801, 1987.
13. Coleman, D. J., Lizzi, F. L., Silverman, R. H., Rondeau, M. J., Smith, M. E., and Torpey, J. H.: Acoustic biopsy as a means for characterization of intraocular tumors. In Henkind, P. (ed.): *Acta. XXIV International Congress of Ophthalmology*, vol. 1. Philadelphia, J. B. Lippincott Company, 1983, pp. 115-118.
14. Feleppa, E. J., Lizzi, F. L., Coleman, D. J., and Yaremko, M. M.: Diagnostic spectrum analysis in ophthalmology. A physical perspective. *Ultrasound Med. Biol.* 12:623, 1986.
15. Coleman, D. J., Silverman, R. H., Rondeau, M. J., Lizzi, F. L., McLean, I. W., and Jakobiec, F. A.: Correlations of acoustic tissue typing of malignant melanoma and histopathologic features as a predictor of death. *Am. J. Ophthalmol.* 110:380, 1990.
16. Kidd, M. N., Lyness, R. W., Patterson, C. C., Johnson, P. B., and Archer, D. B.: Prognostic factors in malignant melanoma of the choroid. A retrospective survey of cases in Northern Ireland between 1965 and 1980. *Trans. Ophthalmol. Soc. U.K.* 105:114, 1986.
17. Cox, D. R.: Regression models and life-tables. *J. R. Stat. Soc.* 34:187, 1972.
18. Therneau, T. M., Grambsch, P. M., and Fleming, T. R.: Martingale-based residuals for survival models. *Biometrika* 77:147, 1990.
19. Schmee, J., and Hahn, G. J.: A simple method for regression-analysis with censored data. *Technometrics* 21:417, 1979.